a

Bucket-strict accuracy (%)

0 20 40 60 80 100

Overall | Yes | Contextual | No

b

Haiku 4.5 (naive) · Haiku 4.5 (NCBI) · Haiku 4.5 (PubMed) · Haiku 4.5 (web) · Sonnet 4.6 (naive) · Sonnet 4.6 (NCBI) · Sonnet 4.6 (PubMed) · Sonnet 4.6 (web) · Opus 4.8 (naive) · Opus 4.8 (NCBI)

Exact-reason accuracy (%)

0 20 40 60 80 100

73%

Overall (n=147) | classical surf rcptr (n=28) | multipass exp loops (n=24) | GPI anchored (n=15) | stable cplx partner (n=1) | stable surf attachment (n=4) | cell-state induced (n=7) | tissue restricted (n=5) | lysosomal exocytosis (n=5) | dual localization (n=10) | cytoplasmic (n=14) | nuclear (n=1) | endomem resident (n=12) | nuclear envelope (n=5) | secreted only (n=7) | inner leaflet anchored (n=8) | pMHC only (n=1)

Haiku 4.5 (web — best) · Sonnet 4.6 (NCBI) · Sonnet 4.6 (PubMed) · Opus 4.8 (NCBI)

Showing 16 of 19 TriageReason enum values (present on bench, n ≥ 1). Absent from bench: extracellular_face_protein, mitochondrial_internal, other.

c

Curator ground_truth_reason

classical surf rcptr · multipass exp loops · GPI anchored · extracell. face (other YES) · stable cplx partner · stable surf attachment · cell-state induced · tissue restricted · lysosomal exocytosis · dual localization · cytoplasmic · nuclear · mito internal · endomem resident · nuclear envelope · secreted only · inner leaflet anchored · pMHC only · other

25 | 1 NRROS | 2 SEZ6, DLK1
1 CLDN6 | 23
15
1
2 | 1 CRISP1 | 1 FN1
1 DLL3 | 5 | 1 SRC
4 | 1 KLK2
1 LAMP3 | 3 | 1 SCARB2
1 STIM1 | 1 TYRP1 | 7 | 1 TGOLN2
11 | 3 BTK, JAK1, AKT2
1
1 APPL1 | 1 IZUMO4 | 10
5
7
8
1

genes: 0 5 10 15 20 25

yes-bucket reasons · contextual-bucket reasons · no-bucket reasons · diagonal (reason agrees)

Agent predicted_reason (Sonnet 4.6 + NCBI)